

# Josiah Couch

(580)641-3030 • josiah.couch@gmail.com • LinkedIn • GitHub • ORCID

## Skills

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Python, Jupyter, Pandas, SciPy, NumPy, PyTorch, Sci-Kit Learn, Git, Mathematica, LaTeX, Energy-based models

## Experience

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### Arnaout Lab, Beth Israel Deaconess Medical Center, Boston, MA

*Postdoctoral Research Fellow*

*August 2022 - Present*

- Applying entropy-related concepts to computer vision datasets in an effort to characterize dataset quality, with an emphasis on medical imaging data (arxiv: 2511.03849, 2505.22813, 2407.15724),
- Thought about normalizing maximum entropy models, with a focus on models for IgH and TCR repertoires (arxiv: 2312.12525)

### Boston College, Chestnut Hill, MA

*Postdoctoral Research Fellow in Computer Science Department*

*January 2021 - August 2022*

- Applied results from discrete statistics and graph alignment to prove bounds on the recoverability of the order of a shuffled chain of pairwise correlated Erdős-Rényi graphs

## Education

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### University of Texas at Austin, Austin, TX

*Ph.D in Physics (3.6 GPA)*

*August 2013 - May 2021*

- Adviser: Willy Fischler
- Field: AdS/CFT
- Research:
  - Published six papers in peer-reviewed journals with 14 collaborators from three institutions (including UT).
  - Gave six external talks, five at conferences, and participated in three additional conferences and schools
  - Awarded 2018 OGS Summer Only Fellowship
  - Summer at ATLAS (experimental physics requirement): Optimized feature selection for a particle physics analysis using simulated data, Python, and the ROOT TMVA boosted decision tree implementation (used through pyROOT)
- Teaching:
  - Courses included graduate quantum mechanics (1 semester), undergraduate quantum mechanics (6 semesters), forensic science (2 semesters), electricity and magnetism (3 semesters), and engineering physics labs (5 semesters)
  - Nominated for the UT Services for Students with Disabilities Appreciation Award in Fall 2016

### Oklahoma State University, Stillwater, Oklahoma

*B.S in Physics and B.S. in Mathematics*

*August 2009 - August 2013*

- 2.5 years of undergraduate research
- Analyzed particle physics data from the DØ and ATLAS experiments using C/C++ and the ROOT package

## Select Publications/Preprints

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- P. Nguyen, J. Couch, R. Bansul, A. Morgan, C. Tam, M. Li, R. Arnaout, R. Arnaout, "Which Similarity-Sensitive Entropy?," [arXiv:2511.03849 [cs.IT]].
- J. Couch, M. Li, R. Arnaout, R. Arnaout, "X-Factor: Quality Is a Dataset-Intrinsic Property," [arXiv:2505.22813 [cs.LG]].
- J. Couch, R. Arnaout, R. Arnaout, "Beyond Size and Class Balance: Alpha as a New Dataset Quality Metric for Deep Learning," [arXiv:2407.15724 [cs.CV]].
- J. Couch, R. Arora, J. Braun, J. Kaplinsky and E. Hill, A. Li, B. Altshul, R. Arnaout, "Scaling Monte-Carlo-Based Inference on Antibody and TCR Repertoires," [arXiv:2312.12525 [q-bio.QM]].
- J. Couch, P. Nguyen, S. Racz, G. Stratis and Y. Zhang, "Possibility of entanglement of purification to be less than half of the reflected entropy," [arXiv:2309.02506 [quant-ph]].